

Yuhan Chen

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Education

Hong Kong University of Science and Technology (Guangzhou), AI Thrust
Bachelor of Artificial Intelligence

Sep. 2023 - June 2027 (*Expected*)

MCGA: 3.52/4.3

Professional courses: Machine Learning (A), Introduction to Data Mining (A), Introduction to Embodied AI (A), Natural Language Processing (A), Python Programming for AI (A), Introduction to AI (A), etc.

Research Experience

Enhancement of TimeSeriesScientist (TSci) Agentic Framework

Jan. 2026 - Present

Supervisor: Prof. Hao Liu, AI Thrust, HKUST(GZ)

- Integrated advanced data augmentation and In-Context Learning capabilities into the agent to enhance the handling of missing data and irregular time series.
- Redesigned the Planner agent by incorporating Chain-of-Thought (CoT) reasoning to significantly improve the interpretability and transparency of the model selection process.
- Enhanced the Reporter agent to synthesize complex statistical diagnostics and ensemble results into intuitive, human-readable narratives.

Building a Hybrid Event Sensor Simulator

June 2025 - Dec. 2025

Supervisor: Prof. Hui Xiong, AI Thrust, HKUST(GZ)

- Systematically collected the scene datasets under zero lux, strong light exposure and high contrast scenes to analysis the noise situation.
- Implemented components of the EVS noise model, including event probability estimation using the Q-function and intensity-to-voltage domain mapping.
- Evaluated simulator realism by fine-tuning state-of-the-art video interpolation and deblurring models (e.g., TimeLens, EFNet) on synthetic data.

High-Speed Motion Analysis in Badminton via Event Camera

June 2025 - Sep. 2025

Supervisor: Prof. Hui Xiong, AI Thrust, HKUST(GZ)

- Adapted state-of-the-art models by integrating their structural merits into a custom-designed framework.
- Modified gradient backpropagation functions in the training pipeline: migrated core numerical computation modules to enable end-to-end gradient differentiability.
- Refined dataset construction and DataLoader architectures to enhance compatibility with diverse input modalities.

Project Experience

Multi-Skill Pipeline for Daily Paper Briefings

Feb. 2026 - May. 2026

Supervisor: Prof. Enyan Dai, AI Thrust, HKUST(GZ)

- Designed a "Orchestrator + Skills" architecture in which skills communicate only through versioned JSON files under shared_data, keeping modules fully decoupled.
- Implemented the unified orchestrator arxiv_agent.py with serial pipeline execution, idempotent re-runs, and fast-fail short-circuit when the data-collector returns zero papers.
- Built the data-collector skill: arXiv paper fetching, metadata extraction, 384-dim MiniLM embedding generation, and construction of the cosine-similarity paper graph and co-author graph consumed by the downstream ranker.

Explanations of LLM — AttnLRP on GPT-2

Mar. 2025 - June. 2025

Supervisor: Prof. Sihong Xie, AI Thrust, HKUST(GZ)

- Set up the lxt AttnLRP framework via an efficient monkey-patch implementation on GPT-2, establishing the code pipeline shared by the whole project.
- Designed three visualization tools which surfaced GPT-2's negative-logit sign-flip artifact in AttnLRP, and co-developed the contrastive-gradient masking fix that restored correct relevance signs.

Brain-to-Image Retrieval & Reconstruction on THINGS-EEG2

Feb. 2026 - May. 2026

Supervisor: Prof. Chen Liang, AI Thrust, HKUST(GZ)

- Designed the Hierarchical Visual Features (HVF) pipeline, fusing CLIP RN50, CLIP ViT-B/32, and the Stable Diffusion VAE into a single regression target for the downstream EEG encoder.
- Replaced plain-sum fusion with a sigmoid-gated fusion (bias init +3.0), letting the model learn to attenuate unhelpful encoders — materially improved Top-5 / Top-10 / Mean Rank.
- added a fourth encoder (DINOv2 ViT-S/14 fine-tuned on Depth Anything V2) to inject a 3D/depth prior missing from purely semantic features.

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Social Navigation with 3D Simulator

Aug. 2025 - Dec. 2025

Supervisor: Prof. Junwei Liang, AI Thrust, HKUST(GZ)

- Developed and optimized social navigation policies for embodied agents in Habitat Lab, with a solid understanding of embodied perception, robot-environment interaction in complex simulated scenes..
- Tuned PPO training pipelines and robot-related configuration parameters, including navigation, sensing, and action settings, to improve policy and task performance.
- Enhanced agent performance significantly by modifying the code in additional metrics modules to incorporate custom penalty mechanisms, including orientation alignment and path overlap penalties.

Competition and Awards

Ascend AI Innovation Contest Season 8 (Atlas 910B)

June 2026

National competition organized by Huawei; awarded Outstanding Prize.

- Implemented the Scale and Atanh operators as custom AI Core kernels in Ascend C, using msopgen-generated op_host (Tiling + InferShape + prototype registration) and op_kernel projects targeting the Atlas 910B (aarch64) platform.
- Designed generalized tiling strategies covering arbitrary rank inputs (1D / 2D / 4D / 6D) with non-32-aligned tail handling, and used dispatch technology to specialize kernels per dtype (fp16 / bf16 / fp32) and shape regime.

Deblurring for HVS/HybridEVS Camera @MIPI-challenge

July 2025

International Competition; Rank 6 over 130 participants (All over the world)

- Revised the gradient backpropagation components within the training framework: migrated core numerical computation modules from NumPy to PyTorch, which established end-to-end gradient differentiability.
- Applied novel deblurring-specific models (EFNet, ESL) to tackle image deblurring tasks, leveraging their specialized architectures to enhance restoration performance for blurred visual content.

2024 Mecha Master University Alliance Competition (Southwest Region)

Feb. 2024

National university-level competition; placed top 8 in the infantry confrontation competition.

- Proficiently acquired SolidWorks skills through self-directed learning, independently designing and modeling mechanical parts and assembly structures.
- Fabricated custom parts using 3D printers and lathes, ensuring precision in physical component production.
- Collaborated closely with the electrical control team to coordinate wiring and electrical integration, resolving technical issues to ensure seamless system functionality.

Skills

- Proficient in Python, C++, and SQL
- Experienced with PyTorch, TensorFlow, and scikit-learn
- Solid understanding of Transformers, attention mechanisms, and prompt engineering
- Experienced with LLM application methods, including CoT, ReAct, and RAG
- Experienced with LLM fine-tuning techniques, including SFT, LoRA, and PEFT
- Comfortable with Linux, Git, and GitHub development environments
- Experience in model integration, optimization, and deployment, as well as open-source project reproduction, adaptation, and engineering implementation
- Proficient in English, with the ability to read academic papers and technical documentation

Extracurricular Activities

- Served as an English-only tutor for 4th-grade primary school students, delivering full English instruction to enhance their language proficiency and academic performance.
- Worked as a teaching assistant at Minxin Hong Kong School, supporting classroom management, student guidance, and educational activity organization.
- Held the position of Organization Committee Member in the Youth League Branch of RC2 Community 3, responsible for organizing league activities, coordinating member participation, and promoting branch development.
- Participated in observational visits to leading enterprises including Yunzhou Intelligence, DJI, and BYD, gaining insights into advanced technologies, industrial operations, and corporate innovation practices.